

Practical 3 – The Transistor as a Switch

Booklet 3, section 11, page 27. Success criteria 5.1, 5.2 and 5.6.

Everything you write goes in your jotter. Copy each numbered table before you start that section, and answer each question by its number.

The idea being tested

A transistor is an electronic switch: a small voltage at its base switches on a much larger current through it.

An NPN transistor switches on when the voltage between its base and its emitter reaches about 0.7 V – the switching voltage.

Check your bench

Your transistor may not look like the one in the picture. Whatever its case, find its base (B), collector (C) and emitter (E) on the card at your bench before you push it in.

3.1 What you are testing

Aim: to find the base-emitter voltage at which the transistor switches the LED on.

5 V supply. An LDR over a variable resistor R makes the input, a 10 k Ω resistor feeds the base, and the LED has a 270 Ω resistor in series.

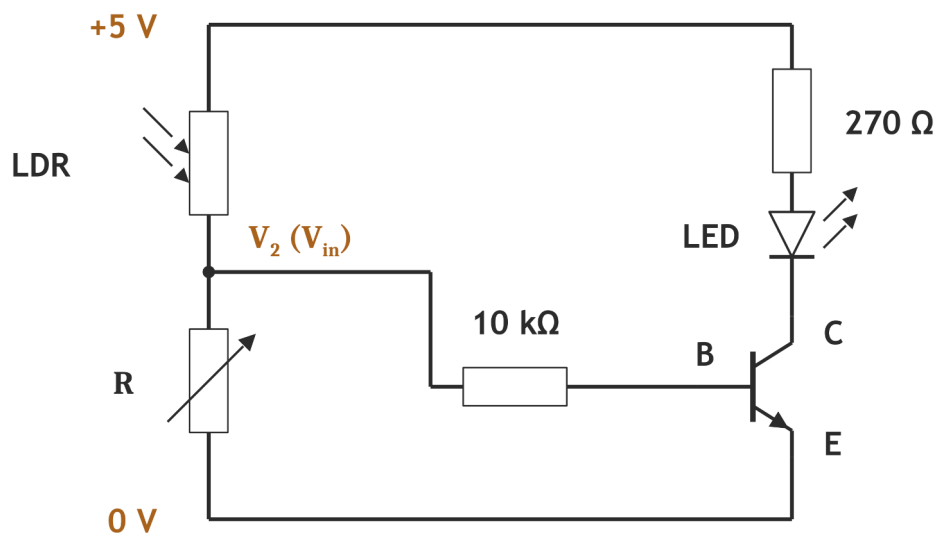
3.2 Predict

Table 3.2 – LED on or off?

Base-emitter voltage (V)	LED on or off?
0.2	
0.4	
0.6	
0.8	

Q1 R is the variable resistor below the LDR. State whether R must be made larger or smaller to switch the LED on, and explain why. [2]

3.3 Simulate in Yenka



Change R a little at a time. Record each reading, and underline the one at which the LED first comes on.

Table 3.3 – simulated readings

Base-emitter voltage (V)	LED on or off?
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Q2 Describe what happened to the LED as the base-emitter voltage rose past the switching voltage. [2]

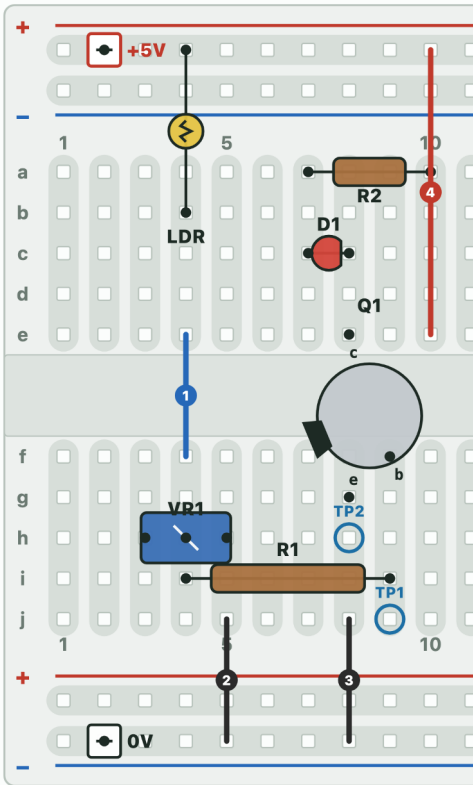
Now turn the LED round in Yenka. Test the circuit, find what is wrong, change it and test it again.

Table 3.3b – testing the circuit

Planned test	Expected result	Initial test result	Planned amendment	Re-test result
Shine light on the LDR.	The LED comes on.			

Q3 Copy Table 3.3b and complete the last three columns. [3]

3.4 Build on the breadboard



Q1 straddles the channel: collector (c) above, base (b) and emitter (e) below.

Turn VR1 slowly. Read the base-emitter voltage between TP1 (base) and TP2 (emitter) at each step, and underline the reading at which the LED first comes on.

Table 3.4 – measured readings

Base-emitter voltage (V)	LED on or off?
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3.5 Compare

Table 3.5 – the switching voltage

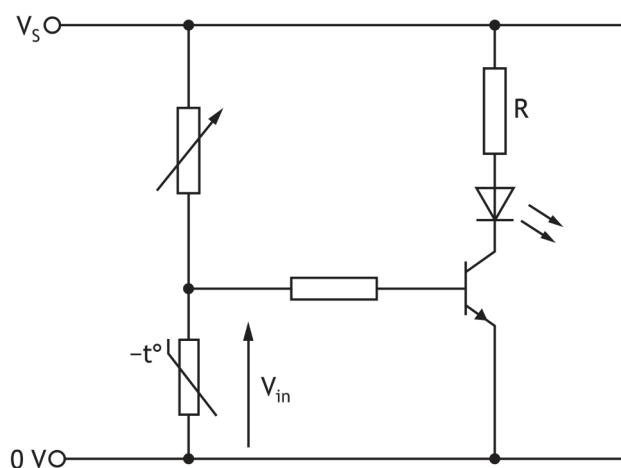
	Predicted	Simulated	Measured
Switching voltage (V)			

Q4 Give one reason why your measured switching voltage is not exactly the simulated one. [1]

3.6 What the results tell you

- Q5 State the function of the transistor in this circuit. [1]
- Q6 Sketch the base resistor and the transistor, and mark with a cross (X) where an ammeter would measure the current into the base. [1]
- Q7 Describe the operation of the circuit as the light level increases. Refer to the resistance of the LDR, V_2 (V_{in}), the transistor and the LED. [4]
- Q8 Draw the circuit changed so that the LED comes on when it gets dark. [2]
- Q9 Explain why the LED has a $270\ \Omega$ resistor in series with it. [2]
- Q10 Explain why the $10\ \text{k}\Omega$ resistor sits between the voltage divider and the base. [2]

From a past exam paper



This circuit is used to indicate when a slow cooker is at the required temperature.

Sources & credits: © Qualifications Scotland (SQA). National 5 Engineering Science 2021, Question 10(c) and (d).

- Q11** Describe the operation of the circuit as the temperature increases. Include reference to the resistance of the thermistor and the voltage V_{in} . [4]
- Q12** Describe the function of resistor R in this circuit. [1]